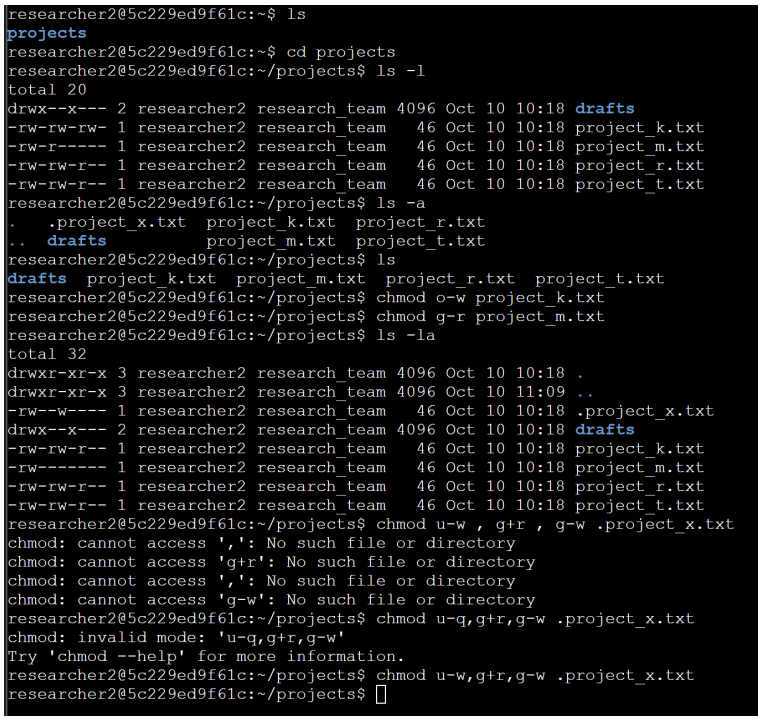
File permissions in Linux

Project description

In this project, I acted as a security professional managing authorizations for a research team. I examined existing file and directory permissions in a Linux environment and modified them using the command line to ensure that access aligned with the organization's strict security policies and the principle of least privilege.

Check file and directory details

To check the permissions of all files and directories, including hidden ones, I used the command: `ls -la` This command displayed the contents of the `projects` directory along with a 10-character string for each item, such as `-rw-rw-rw-` for the `project_k.txt` file or `drwx--x---` for the `drafts` directory.



```
researcher2@5c229ed9f61c:~$ ls
projects
researcher2@5c229ed9f61c:~$ cd projects
researcher2@5c229ed9f61c:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Oct 10 10:18 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Oct 10 10:18 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 10 10:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 10 10:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 10 10:18 project_t.txt
researcher2@5c229ed9f61c:~/projects$ ls -a
.   .project_x.txt  project_k.txt  project_r.txt
..  drafts          project_m.txt  project_t.txt
researcher2@5c229ed9f61c:~/projects$ ls
drafts  project_k.txt  project_m.txt  project_r.txt  project_t.txt
researcher2@5c229ed9f61c:~/projects$ chmod o-w project_k.txt
researcher2@5c229ed9f61c:~/projects$ chmod g-r project_m.txt
researcher2@5c229ed9f61c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 10 10:18 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 10 11:09 ..
-rw--w---- 1 researcher2 research_team   46 Oct 10 10:18 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 10 10:18 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 10 10:18 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 10 10:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 10 10:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 10 10:18 project_t.txt
researcher2@5c229ed9f61c:~/projects$ chmod u-w , g+r , g-w .project_x.txt
chmod: cannot access ',': No such file or directory
chmod: cannot access 'g+r': No such file or directory
chmod: cannot access ',': No such file or directory
chmod: cannot access 'g-w': No such file or directory
researcher2@5c229ed9f61c:~/projects$ chmod u-q,g+r,g-w .project_x.txt
chmod: invalid mode: 'u-q,g+r,g-w'
Try 'chmod --help' for more information.
researcher2@5c229ed9f61c:~/projects$ chmod u-w,g+r,g-w .project_x.txt
researcher2@5c229ed9f61c:~/projects$
```

Describe the permissions string

The 10-character string represents the file type and the specific permissions for the user (owner), group, and others.

- The 1st character indicates the file type (`-` for a regular file, `d` for a directory, and `l` for a symbolic link).
- Characters 2-4 represent the user's permissions (`r` for read, `w` for write, `x` for execute).
- Characters 5-7 represent the group's permissions.
- Characters 8-10 represent the permissions for all other users on the system. For example, `-rw-rw-rw-` means it is a regular file where the user, group, and others all have read and write permissions, but no execute permissions.

Change file permissions

The organization dictates that "others" should not have write access to any files. Based on the initial permissions, `project_k.txt` allowed write access to others. To remove write permissions for others on the `project_k.txt` file, I used the command: `chmod o-w project_k.txt`

Change file permissions on a hidden file

The hidden file `.project_x.txt` needed to be archived. It should not have write permissions for anyone, but the user and group should retain read access. To remove the appropriate write permissions and ensure read access, I modified it using the command: `chmod u-w,g+r,g-w .project_x.txt`

Change directory permissions

Only the user `researcher2` should have access to the `drafts` directory and its contents. To restrict access and remove the execute permissions for the group, I used the command: `chmod g-x drafts`

```
researcher2@5c229ed9f61c:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Oct 10 10:18 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 10 10:18 project_k.txt
-rw------- 1 researcher2 research_team   46 Oct 10 10:18 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 10 10:18 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 10 10:18 project_t.txt
researcher2@5c229ed9f61c:~/projects$ chmod g-u drafts
researcher2@5c229ed9f61c:~/projects$
```

Summary

By utilizing Linux command-line tools such as `ls -la` and `chmod`, I successfully audited and updated the file system permissions for the research team. This ensured that unauthorized users were stripped of excessive access, sensitive hidden files were protected from modification, and private directories were restricted exclusively to their appropriate owners, thereby hardening the system's overall security posture.